

AI-Powered Sentiment Analysis with Secure CI/CD Deployment

Abstract

This project presents an end-to-end implementation of an AI-powered sentiment analysis service, emphasizing production-grade software engineering and deployment practices over standalone model development. The core machine learning component utilizes a Scikit-learn classifier to categorize text (such as reviews or social media posts) into positive, negative, or neutral sentiments. To transition the model from a local script to a scalable service, it is wrapped in a Flask REST API and containerized using Docker, ensuring cross-environment consistency.

The project lifecycle is strictly governed by Agile methodologies, utilizing Scrum frameworks and sprint planning for structured execution. A primary focus is placed on DevSecOps and MLOps through a robust Continuous Integration and Continuous Deployment (CI/CD) pipeline powered by GitHub Actions. This automated workflow builds the containerized application and enforces strict security gates, integrating SonarCloud for code quality analysis and Trivy for container vulnerability scanning.

Upon passing all automated checks, the secure model is deployed to AWS SageMaker as a live, accessible cloud endpoint with foundational monitoring. By seamlessly integrating Agile planning (CO1), DevOps automation (CO2), DevSecOps vulnerability scanning (CO3), and MLOps cloud deployment (CO4), this project demonstrates a comprehensive, industry-standard approach to shipping secure, reliable, and scalable AI products.

Keywords: Machine Learning, Cloud DevOps, CI/CD, DevSecOps, AWS SageMaker, Docker, Agile Methodology, Sentiment Analysis.

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